

Syntax for Functions (Lecture 13)

→ int, char, bool, float, ll, void

The need for return type

```
✓int product(int4 a, int11 b){
    ✓return4*11 a*b;
}

•int main(){
    cout<<product(4,11);
}
```

44

Output

. 44

return ;

→ fⁿ ko khatam

or

line khatam

Function ki khud ki
ek value hoti hai

loop



khatam ?



condition



break

Example

```
1  #include<iostream>
2  using namespace std;
3
4  int sum(int x, int y) {
5      |    return x + y;
6  }
7
8  int main() {
9      |    cout << sum(40, 63);
10 }
--
```

What is `int main()` ?

entry pt. of a c++ code.

Some **inbuilt library** functions

$\min(a, b)$

$\max(a, b)$

$\text{sqrt}(n)$

Q// Find the max of 3 numbers using builtin fn

$\max(a, \max(b, c)) ;$

Ques: Take 'n' and 'r' as input and print nCr and nPr .

$${}^nP_r = \frac{n!}{(n-r)!}$$

$${}^nC_r = \frac{n!}{r! \times (n-r)!}$$

$$\text{Ex: } {}^6C_2 = \frac{6!}{2! \times 4!} = \frac{6 \times 5 \times \cancel{4 \times 3 \times 2 \times 1}}{2 \times 1 \times \cancel{4 \times 3 \times 2 \times 1}} = 15$$

```
1  #include<iostream>
2  using namespace std;
3
4  int fact(int x) {
5      int f = 1;
6      for(int i = 2; i <= x; i++) {
7          f *= i;
8      }
9      return f;
10 }
11
12 int main() {
13     int n;
14     cout << "Enter n : ";
15     cin >> n;
16     int r;
17     cout << "Enter r : ";
18     cin >> r;
19     int nfact = fact(n);
20     int rfact = fact(r);
21     int nrfact = fact(n-r);
22     int ncr = nfact / (rfact * nrfact);
23 }
```

Ques : Pascal Triangle

_ _ _ _ 1
 _ _ _ _ 1 _ 1
 _ _ _ 1 _ 2 _ 1
 _ _ 1 _ 3 _ 3 _ 1
 _ 1 _ 4 _ 6 _ 4 _ 1
 1 _ 5 _ 10 _ 10 _ 5 _ 1

$n = 5$

1
 1 1
 1 2 1
 1 3 3 1
 1 4 6 4 1
 1 5 10 10 5 1

Ques : Pascal Triangle

$n = 5$

	0	1	2	3	4	5	$\rightarrow j$
0	1						
1	1	1					
2	1	2	1				
3	1	3	3	1			
4	1	4	6	4	1		
5	1	5	10	10	5	1	
$\downarrow$							
$\downarrow$							

$$\begin{array}{cccc}
 {}^0C_0 & & & \\
 {}^1C_0 & {}^1C_1 & & \\
 {}^2C_0 & {}^2C_1 & {}^2C_2 & \\
 {}^3C_0 & {}^3C_1 & {}^3C_2 & {}^3C_3 \\
 & \vdots & &
 \end{array}$$

Are arguments passed actually the same ?

```
✓void change(int x){  
|   ✓x = 20;  
| }  
✓int main(){  
|   ✓int x = 10;  
|   ✓change(10);  
|   ✓cout<<x<<endl;  
| }  
|
```

20
~~10~~
x

10
x

Output
10

Formal parameters & Actual parameters

↓
parameters
(variables)

↓
arguments
(values)

Scope of variable (Local Scope)

```
void fun() {
```

```
    int x = 10;
```

```
    cout << x << endl;
```

```
}
```

```
int main() {
```

```
    fun();
```

```
    // cout << x;    // Error
```

```
}
```

Scope of variable (Block Scope)



Earn₹s

```
int main() {  
    if (true) {  
        int x = 5;  
        cout << x << endl;  
    }  
    // cout << x;    // Error  
}
```

Global Variables

```
#include <iostream>
using namespace std;
```

```
int x = 100;
```

```
void fun() {
    cout << x << endl;
}
```

```
int main() {
    cout << x << endl;
    fun();
}
```

Request
↓
Pls Contest do.